

NanoGT Match Report: X-linked myotubular myopathy

Disease: X-linked myotubular myopathy (ORPHA:596)

Primary gene: MTM1

Gene CDS: 1878 bp

Inheritance: X-linked recessive

Target tissues: muscle

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	AT132	AAV8	7.3/10	[YELLOW] Medium	phase3
2	SRP-9001	AAV9	6.8/10	[YELLOW] Medium	approved
3	OAV101-IT	AAV9	6.7/10	[YELLOW] Medium	approved
4	Zolgensma	AAV9	6.7/10	[YELLOW] Medium	approved
5	Strimvelis	LV	6.3/10	[YELLOW] Medium	approved

Match #1: AT132

Precedent disease: X-linked myotubular myopathy

Vector: AAV8

Tissue target: muscle

Composite score: 7.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.80	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	7.28	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1878bp / cargo 4700bp (40% utilized)
- Vector tropism plus precedent target match: muscle
- Both intracellular proteins
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: myopathy
- Approval status: phase3
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Narrow therapeutic window — neonatal onset requires delivery within weeks of birth; trial design must incorporate newborn screening programmes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera

Match #2: SRP-9001

Precedent disease: Duchenne muscular dystrophy

Vector: AAV9

Tissue target: muscle

Composite score: 6.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	0.50	2.0	Same se-creted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.80	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	6.83	10.0	Raw sum / 18 × 10

Rationale

- Gene CDS 1878bp / cargo 4700bp (40% utilized)
- Vector tropism plus precedent target match: muscle
- Protein class mismatch
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: myopathy
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Narrow therapeutic window — neonatal onset requires delivery within weeks of birth; trial design must incorporate newborn screening programmes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera

Match #3: OAV101-IT

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/spinal cord

Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.80	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	6.67	10.0	Raw sum / 18×10

Rationale

- Gene CDS 1878bp / cargo 4700bp (40% utilized)
- Vector tropism overlaps muscle, but precedent target is CNS/spinal cord
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Pathway match: motor_neuron
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Narrow therapeutic window — neonatal onset requires delivery within weeks of birth; trial design must incorporate newborn screening programmes

- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera

Match #4: Zolgensma

Precedent disease: Spinal Muscular Atrophy

Vector: AAV9

Tissue target: CNS/motor neuron

Composite score: 6.7 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	1.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.00	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.80	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	6.67	10.0	Raw sum / 18×10

Rationale

- Gene CDS 1878bp / cargo 4700bp (40% utilized)
- Vector tropism overlaps muscle, but precedent target is CNS/motor neuron
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Pathway match: motor_neuron
- Approval status: approved
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Narrow therapeutic window — neonatal onset requires delivery within weeks of birth; trial design must incorporate newborn screening programmes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera

Match #5: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 6.3 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	1.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	0.70	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.80	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.60	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.90	1.0	Established delivery route to target tissue
TOTAL (normalised)	6.28	10.0	Raw sum / 18×10

Rationale

- Gene CDS 1878bp / cargo 8000bp (23% utilized)
- No tissue overlap (disease: ['muscle'], vector: ['hematopoietic'])
- Both intracellular proteins
- LOF inheritance — compatible for gene replacement
- Different pathway (myopathy vs immune_hematopoietic)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Narrow therapeutic window — neonatal onset requires delivery within weeks of birth; trial design must incorporate newborn screening programmes
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: moderate immune surveillance — standard immunosuppression protocols typically sufficient
- Promoter availability: MHCK7, CK8, Desmin — validated in Elevidys (DMD) and SMA programs
- Route of administration: IV systemic or intramuscular injection — well established; used in SMA, DMD, Glybera